

Introduction to DevOps

Final Project

- You will work in groups of 4-5s
- Reminder – the project is 50% from the final grade
- The project needs to be sent to me **till 15/06/26** midnight (after that date, your grade will get down)

Project

Background: You're a DevOps Engineer working for "MeTA" corporate.

Your mission is to deliver your application from development into production, leveraging the knowledge you gained in DevOps course:

Orchestrate, control, deploy, automate, monitor and performance test / analysis. The below should be as a single CI/CD pipeline.

Here's the scenario:

1. Create/find a simple web application (jsp) with at least one link, one button and one input text box
2. Store the code in GIT and GitHub
3. Create a folder in Tomcat under webapps that its folder name includes your names
4. Deploy the simple web application into production environment - the Tomcat (Hint: use Jenkins job to take your code from Git/GITHub into the Tomcat)
5. **Bonus – expose your application to the internet by providing a public IP, perform bullets 6 – 10 accordingly, without compromise on good results / graphs! +10 points bonus**
6. Create **availability monitor** to the application, you can use "SiteMonitorLite" or "UptimeRobot" You should trigger it via Jenkins job to work every 5 minutes
7. Automate a simple test with **5 validations** of your application functionality, using Selenium IDE or similar. You should trigger it via Jenkins job
8. Use Gatling to find your application **max limit**
9. Perform 5 minutes **Load** test on your app with Gatling, trigger it via Jenkins job
10. Perform 5 minutes of **Stress** test on your app with Gatling, trigger it via Jenkins job

I.E. – you can get up to project grade of 110!!!

Finished? Great!! Please send me the following:

- a) The jsp file you used
- b) Screenshot of your Github with the application/jsp in it
- c) Screenshot of the application in your tomcat (I'd like to see the url – localhost:8080/...)
- d) A link to your github public repo
- e) Name of monitor tool you used, what did you monitor and a screenshot of your monitor passed
- f) Selenium IDE file (.side)
- g) Selenium IDE screenshot of passed run, explain what kind of validation you used and why (positive test, negative test, use verify and assert in a smart way). Specify in words what validations you used and why
- h) Describe your HAR scenario in words (for example: click on... -> type... -> click ...)
- i) Attach the HAR file
- j) Write in the email what is your app max limit, explain why this is the limit and how you found it

- k) Attach 3 screenshots of: max limit, load and stress gatling run summary (screenshot of the CMD)
- l) Attach 3 PDFs (max limit, load and stress) with Gatling results/graphs (export/print the index.html to pdf), explain why the results look this way. Guess, what happened in the system

Send those 12 items to mosh.mta2@gmail.com the email subject: "Final Exercise from: <yournames>"

Project defense instructions

- A. You will need to demo your work from your Laptop connected to the projector/ google meet
- B. Every project defense – 15-20 minutes
- C. You will be asked in live to do a small code change in Git and push to GitHub, be prepared.
- D. Predefined Jenkins job will identify the new code and copy it to Tomcat (production), same as step 4 above
- E. You will perform also steps 6-10
- F. During the defense – you will be asked to answer questions related to the topics learned in the class
- G. The defense is part of your grade

Appendix

- * You can find examples for index.jsp files on the internet, or chatGPT
- ** You can use whatever monitor tool or application you want
- *** Many ways to run selenium from Jenkins at least 10 different ways!
- **** Gatling documentation: <https://gatling.io/docs/gatling/reference/current/cheat-sheet/>

Reminder: Install plugin in Jenkins: Manage Jenkins->Manage Plugins->Advanced->upload plugin

